Mohammad Amara

Computer Science and Embedded Systems Engineering Student

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Summary

Engineering student at ENSSAT (Université de Rennes) in computer science and embedded systems. Hands-on experience in FPGA design with VHDL, C programming on ARM Cortex-M microcontrollers, Linux servers, and full-stack web and mobile development. Lead developer of Solibox, an offline education server on Raspberry Pi that won the 2026 PIEED award and was featured in Ouest-France.

Education

Engineering degree (Master's level), computer science and embedded systems
ENSSAT, Université de Rennes, Lannion, France 2024 to present

- French diplôme d'ingénieur, equivalent to a Master's. Options: hardware design, signal processing, embedded systems, Internet of Things, artificial intelligence, cybersecurity.

CPGE MPSI / MP, mathematics and physics
Lycée Technique, Mohammedia, Morocco 2022 to 2024

- Two-year intensive program preparing for the engineering school entrance exams.

Experience

Lead developer and technical coordinator (volunteer), Solibox
Togo Vert et Vivant Trégor (non-profit), Brittany, France Mar 2025 to present

- Designed and built an offline education server on Raspberry Pi: its own WiFi network with a captive portal giving access to PDFs, quizzes, an offline copy of Wikipedia and videos without internet.
- Flask web application behind NGINX (Gunicorn, systemd) with PostgreSQL for newsletter subscribers, quizzes with user accounts and an admin dashboard.
- Won the 2026 PIEED student initiatives award (Étudiants et Développement). Featured in Ouest-France.

Event host
City One and Carré Magique (performing arts venue), Lannion, France Oct 2025 to present

- City One (since Oct 2025): ticket scanning, crowd flow, seating and cloakroom in a 2,000-seat venue.
- Carré Magique (from Nov 2026): welcoming audiences, checking tickets, monitoring the venue.

Web development intern (remote)
One-month internship Aug 2026

- Maintained and updated a website with Framer, React, HTML, CSS and JavaScript.

Student library assistant
ENSSAT library, Lannion, France Sep 2025 to Jun 2026

- Loans, returns, shelving and helping users, with fixed hours alongside a full-time engineering course load.

Projects

Neurolith, crypto token analytics platform
Personal project, neurolith.app In progress

- Web app and Discord/Telegram bot: market data through APIs, large language model analysis and a probabilistic decision model that outputs buy, sell or hold signals.
- Social signal module (X, Telegram) in design, local model hosting considered for latency.

Lucent, a modern desktop environment for Linux
Personal project, github.com/MasterKPG/lucent Oct 2026 to present

- Aims to give Linux a new, coherent interface with a glassmorphism design: menu bar, Dock, Control Center, Spotlight, notifications and System Settings for Debian 13.
- Glass made by the ShojiWM Wayland compositor (built from source in Rust, configured in TypeScript) with a Quickshell (QML) shell; C helpers, GLSL effects, full git history.

Sequential FIR filter (systolic project), VHDL and Quartus
ENSSAT, second year Oct 2026, in progress

- 13-tap FIR filter around one multiply-accumulate unit reused 13 times: address generator, circular buffer, coefficient ROM, Moore FSM, testbench with a reference model. Area versus throughput.

Saturating accumulator on FPGA, VHDL
ENSSAT, Digital Systems track May 2026

- 3-bit inputs, 4-bit sum saturating at 15 with an overflow flag, driven by a 5-state machine.
- Verified with 10 test cases in ModelSim, then on a DE10-Lite board; fixed two integration defects invisible in simulation (I/O standard on one pin, reset polarity).

Arduino Uno clone and Pong game
ENSSAT, class project Apr 2026

- Designed an Arduino Uno compatible board (ATmega328P) in EAGLE on my own: schematic, routing, soldering and bootloader.
- Two-player Pong with a classmate on an 8×8 LED matrix with audio output, on that board.

Gate controller, embedded software in C
ENSSAT Jan 2026

- Two-leaf motorised gate on an LPC1768: 13-state machine with a limit switch per leaf, obstacle safety that stops both motors, and automatic closing after 20 seconds.

Smart Garden, microcontroller project in C (team of 4)
ENSSAT, first year 2025 to 2026

- LPC1768 (ARM Cortex-M3) measuring air temperature and humidity, soil humidity and light, with adjustable thresholds, buzzer alarm, LCD status and timer interrupts.
- Password-protected Bluetooth (UART) interface; settings saved in Flash memory.

Other projects
github.com/MasterKPG

- Multi-process parking simulator in C with the POSIX API (FIFOs, shared memory, semaphores).
- AgriMarket: agricultural marketplace for Morocco (Next.js, React Native/Expo, Supabase).
- Half / Half: Roblox precision duel game, asynchronous architecture, full design document.
- Project management coursework (Ada Lovelace challenge): risk analysis with a criticality matrix.

Technical Skills

Programming	C, VHDL, Python (NumPy, SciPy, Pandas, Matplotlib, Flask, Pillow), Java, JavaScript, TypeScript, Rust, HTML and CSS
Hardware	FPGA (DE10-Lite), Quartus, ModelSim, ARM Cortex-M3 (LPC1768), Arduino, EAGLE, ADC, interrupts, UART and Bluetooth
Systems	Linux (Ubuntu, Debian), POSIX API (processes, threads, IPC), Raspberry Pi, NGINX, Gunicorn, PostgreSQL, Git and GitHub
Web and mobile	Next.js, React, React Native/Expo, Supabase, Framer

Languages and Interests

Languages: French, English, Arabic (all bilingual)
Interests: Sailing (Laser Standard), basketball, football